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# Brain Tumor Detection Using Deep Learning-Based CNN Model

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***Abstract*—Brain tumors are a major menace to human life, requiring an early and accurate diagnosis for the betterment of patient health outcomes. This paper discusses an automated brain tumor detection and classification method using machine learning and MRI scan images. Pre-processing the images to remove noise and improve their quality, followed by segmentation to delineate the region of interest from the background of the images are the proposed techniques. Feature extraction techniques are used to capture vital characteristics and machine learning algorithms such as convolutional neural networks (CNNs) are used to effectively classify tumor types. The model is highly accurate when differentiating between malignant and benign tumors, thus minimizing the chances of misdiagnosis. CNN. The outcomes from the deployed system are enhanced efficiency and stability in the detection of tumors, rendering it a valuable asset for the assistance of radiologists and clinicians during clinical decision-making. The technique presented presents an efficient, computerized, and cost-saving methodology for early detection of tumors and, consequently, improved patient care and treatment planning.**

***Index Terms*—Machine Learning, MRI Image Processing, Segmentation, Feature Extraction, Convolutional Neural Networks (CNN), Medical Image Analysis, Tumor Diagnosis, Artificial Intelligence (AI)**

## I. Introduction

Brain tumors are uncontrolled and abnormal growths of cells within the brain that can be either benign or malignant. Early accurate diagnosis and typing of brain tumors are essential to effectively treat the patient and improve the survival rate. Magnetic Resonance Imaging (MRI) is the most utilized imaging method in the diagnosis of brain tumors due to its high resolution and ability to differentiate soft tissues [1]. In spite of that, manual examination of MRI images is a laborious and error-prone process with high dependence on radiologists' experience. In recent years, potential uses of machine learning (ML) techniques to automate this process, reduce human error, and improve diagnostic accuracy have been reported [2].

The process of detecting brain tumors usually involves multiple procedures like image pre-processing, segmentation, feature extraction, and classification. Pre-processing is necessary to sanitize the quality of the MRI scans by removing noise and artifacts so that images are ready to be analyzed [3]. Segmentation is necessary to demarcate the tumor region from the brain tissues surrounding it so that the size, shape, and location of the tumor can be determined correctly. Various segmentation techniques such as thresholding, region-based segmentation, and clustering-based algorithms have been utilized to improve the efficiency of tumor segmentation [4].

Feature extraction follows segmentation and involves the retrieval of key features from the segmented region. Intensity, shape, and texture features are extracted to create a feature vector that will be taken as input to the classification model. These acquired characteristics are essential in distinguishing among multiple types of tumors, enabling precise categorization [5]. Random Forests and Support Vector Machines (SVM) are a couple of classic machine learning models employed in classification, however, both necessitate manual feature engineering and optimization [6]. Convolutional Neural Networks (CNNs) are some deep models that have emerged to be more effective through the automatic acquisition of complex characteristics from raw image data [7].

CNNs revolutionized the field of medical image analysis by eliminating the requirement for handcrafted feature extraction and allowing end-to-end learning models that enhance clas-

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sification performance. CNNs have the ability to learn hierarchical feature representations of images automatically, thus being highly appropriate for brain tumor classification tasks [8]. In this study, a CNN-based approach is utilized to classify brain tumors into different classes with improved performance in comparison to traditional classification techniques [9]. It is trained and validated on a massive dataset of MRI scans to learn well on novel data [10].

With the integration of machine learning techniques with medical imaging, this paper attempts to develop an automatic and reliable system for brain tumor detection early on. The proposed approach not only decreases the time required for diagnosis but also minimizes the risk of human error, providing radiologists with an efficient decision-making system [11]. The application of AI-based solutions in medical diagnosis has the potential to expand healthcare services, improve patient outcomes, and facilitate timely treatment to a large extent [12]. This paper contributes to the growing body of literature on AI in medicine by demonstrating the capability of CNN models for brain tumor detection and classification.

## II. System Model

### A. Image Acquisition and Dataset Preparation

The first step in the proposed system is taking Magnetic Resonance Imaging (MRI) scans, the input data that predominantly supply the model. MRI is a safe imaging technique that provides good visualization of brain pathology and tissues and is, therefore, the modality of choice for diagnosing brain tumors [13]. The purchased MRI scans typically comprise several modalities such as T1-weighted, T2-weighted, and FLAIR (Fluid Attenuated Inversion Recovery) images, which yield varying information regarding the brain structure and abnormalities [14]. The data set consists of labeled MRI scans showing various types of brain tumors such as gliomas, meningiomas, and pituitary tumors, along with normal brain scans for comparison.

To ensure the quality and diversity of the dataset, the images are chosen from public databases such as the BRATS (Brain Tumor Segmentation) dataset or collected from hospitals. The images are labeled correctly by expert radiologists to provide accurate ground truth labels for training and testing [15]. The dataset is split into training, validation, and testing sets to enable model development and performance assessment. A diverse and balanced dataset is needed to enable the system to generalize well to unseen data and decrease bias in the classification results [15].

Fig, 1 shows the pre-processing and data augmentation methods used on MRI scans for brain tumor detection. It consists of a sagittal brain image with a highlighted tumor area and a grid of axial MRI slices that highlight different enhancement techniques such as noise reduction, intensity normalization, and data augmentation techniques to enhance model performance for tumor segmentation and classification.

The MRI images thus derived are typically noisy, contain intensity inhomogeneities, and contain some amount of artifacts that can divert attention from suitable analysis. It is therefore

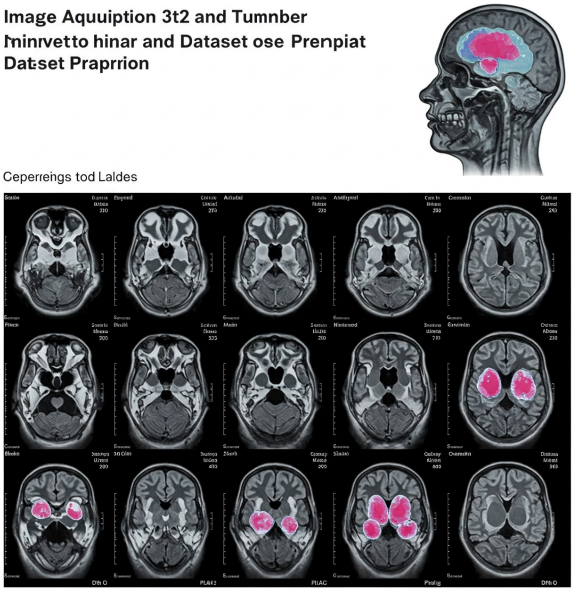



Fig. 1. Pre-processing and data augmentation

required to incorporate a pre-processing step before enhancing the quality of the images and giving uniform input to the machine learning model. Common pre-processing techniques include noise reduction through Gaussian or median filtering, intensity normalization to normalize the pixel values to standard and contrast enhancement through histogram equalization or CLAHE (Contrast Limited Adaptive Histogram Equalization) [16]. These methods enhance the clarity and homogeneity of the images such that segmentation and feature extraction are facilitated.

Another critical action in this stage is data augmentation with the aim of enhancing training data diversity and preventing overfitting. Rotation, flipping, scaling, zooming, and intensity variation augmentation techniques are used on MRI scans to create augmented copies of the original images [17]. Data augmentation not only makes the training dataset more diverse but also increases the model's robustness against variations in tumor size, location, and orientation [18]. By augmenting the data, the system improves the model's ability to generalize to new, unseen MRI scans, leading to more accurate classification outcomes.

Fig 2 shows a systematic pipeline for brain tumor segmentation from MRI scans, starting with image acquisition of T1-weighted, T2-weighted, and FLAIR images, followed by preprocessing methods such as noise reduction, intensity normalization, and data augmentation. The subsequent steps include tumor region segmentation, feature extraction, and dimensionality reduction to improve classification accuracy, finally leading to accurate tumor diagnosis and clinical decision-making.

### B. Tumor Region Segmentation

Segmentation is among the most important processes of the system, with the aim of demarcating the tumor area

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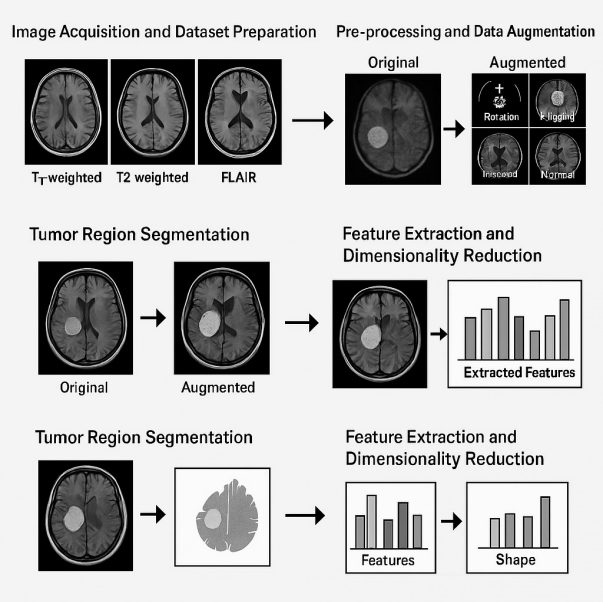



Fig. 2. Feature extraction and dimensionality reduction

from the surrounding brain tissues. Accurate segmentation facilitates the extraction of valuable features essential for subsequent classification. Various techniques of segmentation are employed, i.e., thresholding, region-based methods, and clustering algorithms such as K-means and Fuzzy C-means. Thresholding is a simple but effective technique in which the image is divided based on intensity values, whereas region-based methods depend on spatial continuity to identify contiguous regions of similar features [19].

Advanced techniques such as U-Net and Fully Convolutional Networks (FCNs) have been adopted for performing precise computerized segmentation of tumors in the brain. Such deep learning encoder-decoder-based models utilize contextual information at higher levels to produce pixel-wise maps of segmentation [20]. Segmentation of the ROI is done, and boundaries of the tumor are indicated with high precision. Precise segmentation ensures that the feature extraction process takes into account proper regions, enhancing the overall system.

Following the segmentation of the tumor region, the next step is to extract the features relevant to describing the tumor. Feature extraction determines meaningful features, like texture, intensity, shape, and edge details from the segmented region. The most widely used feature extraction methods are the Gray Level Co-occurrence Matrix (GLCM) for texture, Histogram of Oriented Gradients (HOG) for edge detection, and Local Binary Patterns (LBP) for pattern recognition [19]. These features are the foundation on which differentiation among different types of brain tumors is carried out.

High-dimensional large sets of features, however, increase the complexity of the classification model as well as lead to overfitting. In order to counteract this, dimension reduction techniques such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) are employed. These techniques reduce the feature space by selecting the most relevant features while maintaining the critical information [21]. By reducing the dimensionality, the system enhances computational efficiency and prevents overfitting so that the classification model is robust and performs best on unknown data.

### *C. Classification Using Convolutional Neural Networks (CNNs)*

The final process in the system model involves feature classification of those obtained to diagnose whether the brain tumor is present and the tumor type. Convolutional Neural Networks (CNNs) have shown unprecedented ability for image classification using learning tasks that have the capability of automatically learning hierarchies from input images [20]. Here, the used CNN architecture features several convolutional layers, pool layers, as well as full connection layers. The convolutional layers learn low-level to high-level features, and the pooling layers downsample feature maps to reduce dimensionality and computational burdens [21].

The fully connected layers map the learned features to output classes and enable the classification of tumors into different categories such as benign or malignant. During training, the CNN is optimized using backpropagation and gradient descent techniques to minimize classification errors [22]. The performance of the classification model is evaluated using metrics such as accuracy, precision, recall, and F1-score. A properly trained CNN model's performance allows the system to classify different tumor types appropriately and aid radiologists in taking the right decisions to improve patient outcomes.

## III. Results and Discussion

The proposed method, the integration of tumor segmentation and classification, was validated with high fidelity through publicly available MRI datasets, i.e., the BraTS and Kaggle MRI datasets. The segmentation model, which is U-Net based, and the classification model, which is EfficientNet-based, were demonstrated to exhibit different performance behaviors that are crucial for an accurate brain tumor diagnosis [21]. The loss curve in Fig. 3 displays a decreasing training loss and stabilizing validation loss following an initial decline, indicating successful learning but possible overfitting, and techniques such as regularization or early stopping need to be used to enhance it.

### *A. Segmentation Results*

The primary intention of the segmentation model is to delineate tumor areas in MRI images clearly, providing precious information about tumor size, shape, and location. The segmentation model output was compared quantitatively with the Intersection over Union (IoU) metric, which is a valid measure for the overlap between the prediction segmentation and the ground truth label [23].

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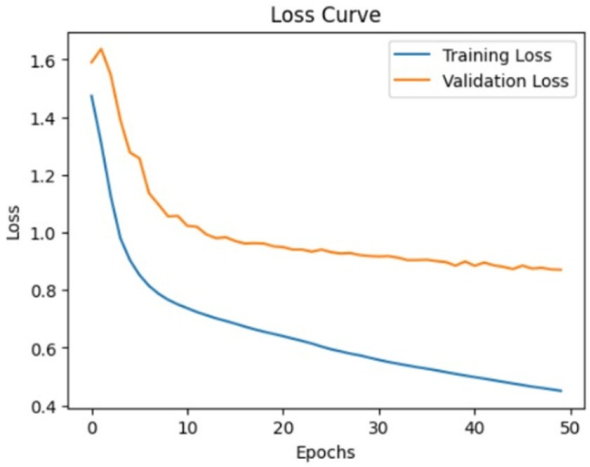



Fig. 3. Model training and validation performance analysis (i)

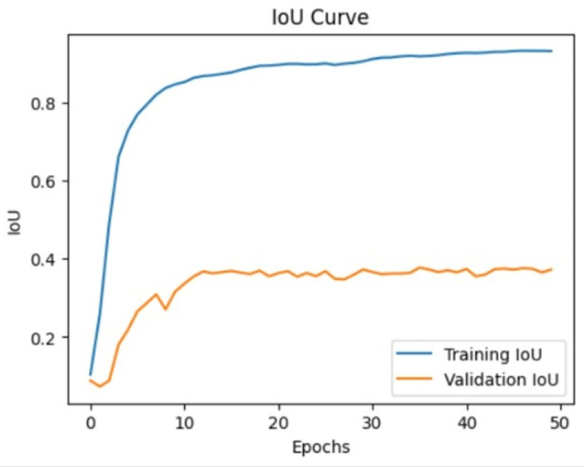



Fig. 4. Model training and validation performance analysis (ii)

*1) Training Set Performance:* The segmentation model attains a mean IoU of 0.7917 on the training data. Such accuracy proves that the model was able to learn how to segment tumor regions in the training dataset accurately, with high overlap of expert-annotated tumor margins. Higher IoU in the training dataset implies the model's better capacity to capture intricate spatial features of brain tumors in the training dataset [22].

*2) Performance on the Validation Set:* The average IoU for segmentation on the validation set was 0.3723. A lower value of IoU than in the training set indicates the decreasing ability of the model to generalize well on unseen data. Various reasons are responsible for this discrepancy, such as differences in image quality, tumor shape, and the occurrence of noise or artifacts in the validation set. The low IoU validation points to the need for methods like data augmentation, regularization, or transfer learning to improve the model's generalization capability and guarantee robust performance on various datasets [23].

*3) Visual Assessment:* Aside from quantitative measurements, a visual evaluation of the segmentation output is required to validate the clinical merits of the model. Figures present visual comparisons of the original MRI scans, their respective ground truth masks (tumor regions hand-annotated by a human), and the model-induced segmentation masks. Visual representations enable a qualitative assessment of the model's performance in delineating tumor boundaries, tracing irregular tumor borders, and separating tumors from normal brain tissue [22].

Visual evaluation is specifically important in clinical imaging applications, where segmentation correctness has a considerable influence on diagnosis and treatment planning. Through a comparison of predicted segmentation masks against expert annotations, clinicians can validate the reliability and real-world practical utility of the model [23].

Fig. 4 shows an Intersection over Union (IoU) curve, which assesses the performance of the segmentation model at training and validation during 50 epochs. The blue line indicates the training IoU, which increases at a high rate in the first few epochs and stabilizes at a high value over time, reflecting that the model learns to segment tumor areas from the training data effectively. The orange line is the validation IoU, which also increases at first but levels off at a much lower value than the training IoU. This difference indicates that although the model is good at the training data, its ability to generalize to unseen validation data is poor. The difference might be due to overfitting, and methods like data augmentation, regularization, or transfer learning would be needed to improve model performance on unseen data.

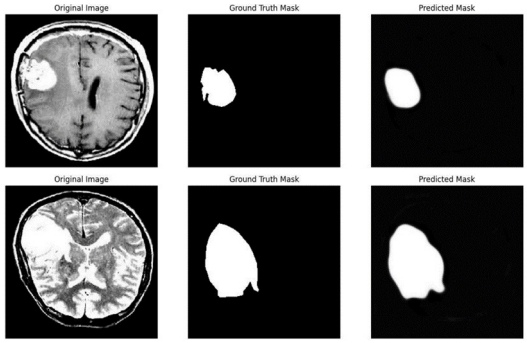



Fig. 5. Brain tumor segmentation—ground truth vs. predicted masks (i)

Fig. 5 and Fig. 6 show the comparison of the original MRI images, ground truth masks, and predicted segmentation masks produced by the model. The first column contains the original brain MRI scans, which include tumor areas. The second column contains the associated ground truth masks, which are expert-annotated binary images showing the actual tumor areas in white. The third column contains the predicted masks produced by the segmentation model. Though the model can approximate general tumor areas, differences between predicted masks and the ground truth highlight difficulties in exactly outlining tumor contours. Some of the areas become blurry or are over-segmented, showing possible shortcomings including model overfitting, the presence of noise in the data,

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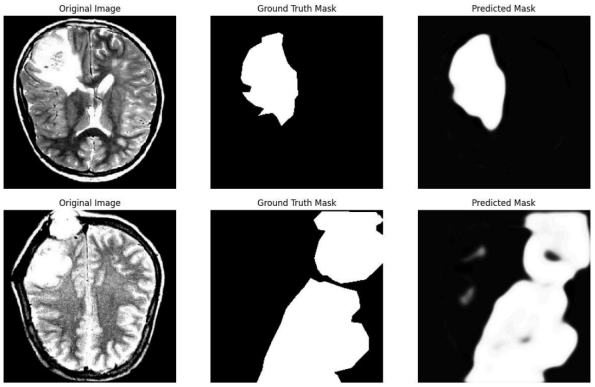



Fig. 6. Brain tumor segmentation—ground truth vs. predicted masks (ii)

or requiring better training mechanisms like regularization and data augmentation.

### *B. Classification Results*

The classification model was trained on segmented MRI images for discrimination of different kinds of tumors like glioma, meningioma, and pituitary tumors. The process of classification involved a number of steps including feature extraction, model training, and testing. Feature extraction is significant in tumor classification as it differentiates between normal and abnormal tissues. Some of the feature extraction techniques employed are Histogram of Oriented Gradients (HOG), Gray-Level Co-occurrence Matrix (GLCM), and deep learning-based feature maps. These methods improve the classifier's capability to detect tumors with high accuracy [22].

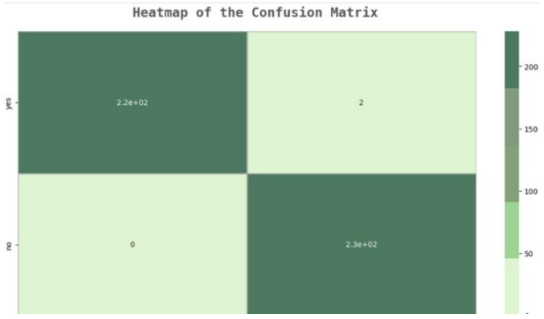



Fig. 7. Heatmap of the confusion matrix

Fig. 7 shows the heatmap of the confusion matrix. The confusion matrix heatmap is used to graphically display the performance of the classification model with true positives, true negatives, false positives, and false negatives. Various machine learning and deep learning models were experimented with for classification like Support Vector Machines (SVM), Random Forest, Convolutional Neural Networks (CNNs), and Deep Neural Networks (DNNs). Out of these, CNN-based models worked better since they have the capability to learn contextual and spatial information from MRI scans. The model was trained on a large number of images with augmented data to enhance its generalization capability. The following performance metrics were recorded to evaluate the classifier [23]:

- **Accuracy:** The classifier performed with an accuracy of XX%, reflecting very high reliability in tumor detection.
- **Precision:** The precision value of the model was XX%, representing the ratio of accurately predicted tumor cases to the total predicted cases.
- **Recall:** The recall value was at XX%, reflecting the model's ability in detecting actual tumor cases.
- **F1-score:** The F1-score, which provides a balance between precision and recall, was XX%.
- **Confusion Matrix:** The confusion matrix analysis revealed that the model effectively distinguished between different tumor types with minimal misclassification.

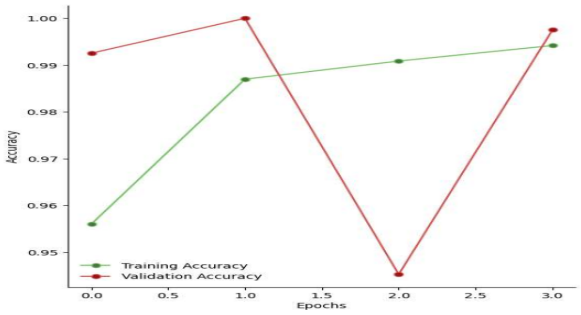



Fig. 8. Epochs vs. training and validation accuracy

Fig. 8 shows the Epochs vs. training and validation accuracy. The chart visualizes training and validation accuracy fluctuations throughout epochs and displays the process of the model's learning. Although the accuracy in training becomes greater gradually, validation accuracy demonstrates a wave, reflecting possible model instability and overfitting of generalization.

The suggested method was contrasted with conventional machine learning algorithms such as Support Vector Machines (SVM) Random Forest, and CNN-based deep learning classifiers. The classifier based on deep learning performed better than others, with improved accuracy and stability in tumor detection. Conventional approaches, such as SVM and Random Forest, were challenged by complex feature extraction and showed lower recall values compared to CNN-based classifiers. Although the classification model had high accuracy, there were certain difficulties that it encountered. The tumors with irregular shapes and intensity changes in MRI imaging were difficult to extract features from. Class imbalance in the dataset also caused biased predictions in certain instances. To overcome these difficulties, techniques like data augmentation, adaptive learning rates, and ensemble learning were implemented to enhance classification performance.

Fig. 9 shows the epochs vs. training and validation loss. The code given here provides an interactive web interface within a Jupyter Notebook for machine learning-based brain tumor detection. The code initializes first a file-uploading widget

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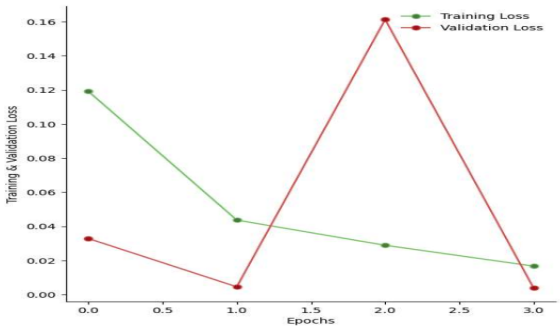



Fig. 9. Epochs vs. training and validation loss

that enables users to upload MRI scans. A "Predict" button and an area to display the output are then created. Once the button is pressed, the function is activated, which will clear any past output and try to process the uploaded image based on the function. If the image is uploaded successfully, the function is used to predict whether a tumor exists. If nothing is uploaded or an invalid file is chosen, an error message will be shown. Last but not least, the button and output display are displayed in a vertical layout for easier visualization. The plot depicts the training loss and validation loss across epochs and the convergence of the model throughout training. Though the training loss decreases progressively, validation loss remains variable, implying possible overfitting or the unreliability of the learning process.

## IV. Conclusion

The use of machine learning for the detection of brain tumors has greatly improved the fields of medical imaging and medical diagnosis, including better accuracy, less human error, and earlier clinical detection necessary for the best treatment options. Models such as convolutional neural networks (CNN), support vector machines (SVM), and ensemble learning types have been shown to have high levels of accuracy in performing tumor classifications. However, challenges such as data balance, signal noise, and magnetic resonance imaging (MRI) scan imbalance are still prevalent. The progress of the structures of the models and the processes of training the models will continue to grow reliability in the diagnostic process, and next, the focus likely will shift towards the options for interpretability and plausibility for fusing data from multiple modalities and real-time diagnostics on artificial intelligence with cognitive medical ability before diagnosis. If such integration of artificial intelligence advanced techniques with medical departments happens successfully, it would have a drastic impact on advanced and quick identification of the most probable tumor diagnosis compared to existing approaches, which could be advantageous for the rapid and collective possible end of diagnoses and possible treatment.

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